Probe design

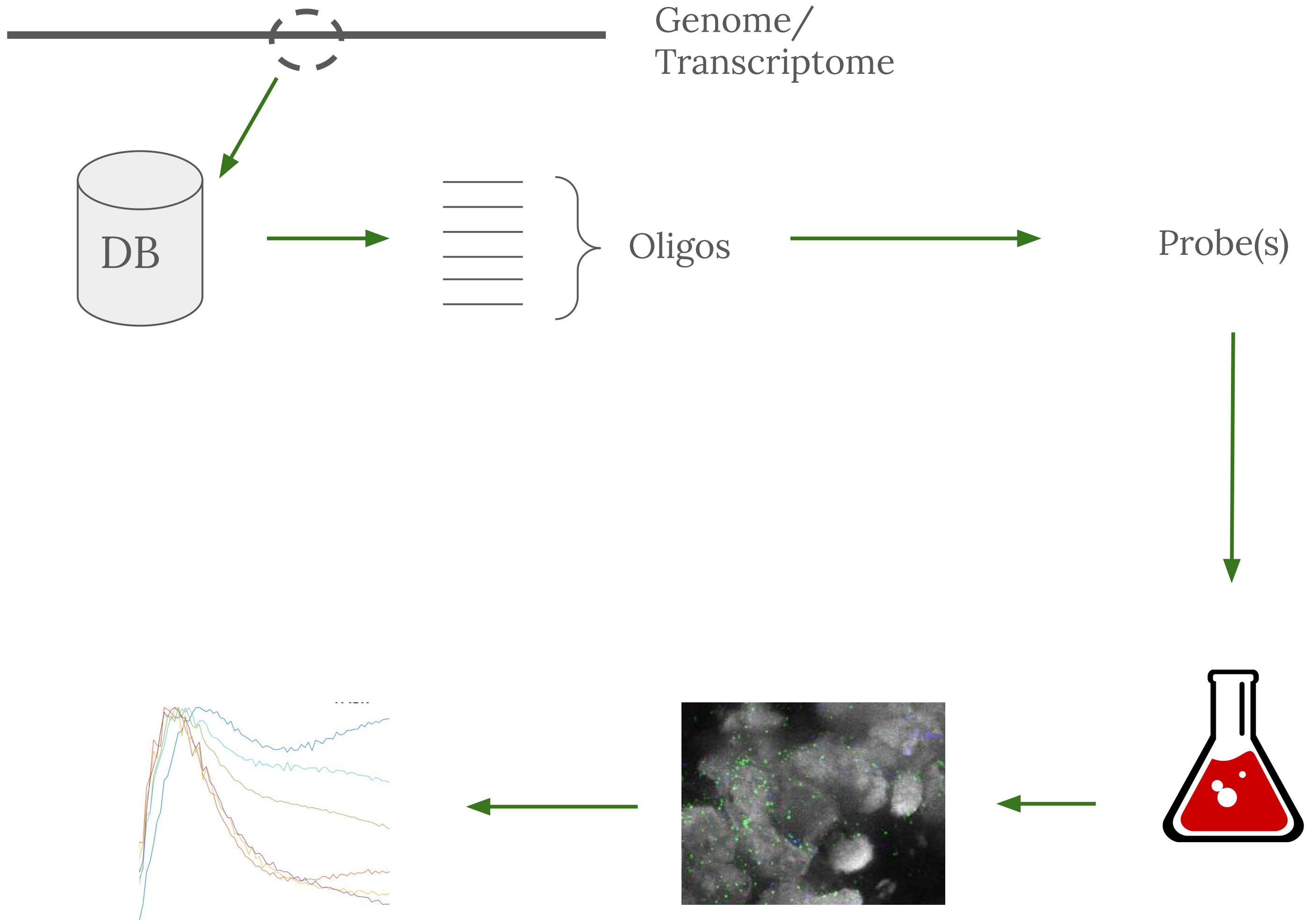
Part I: Theory

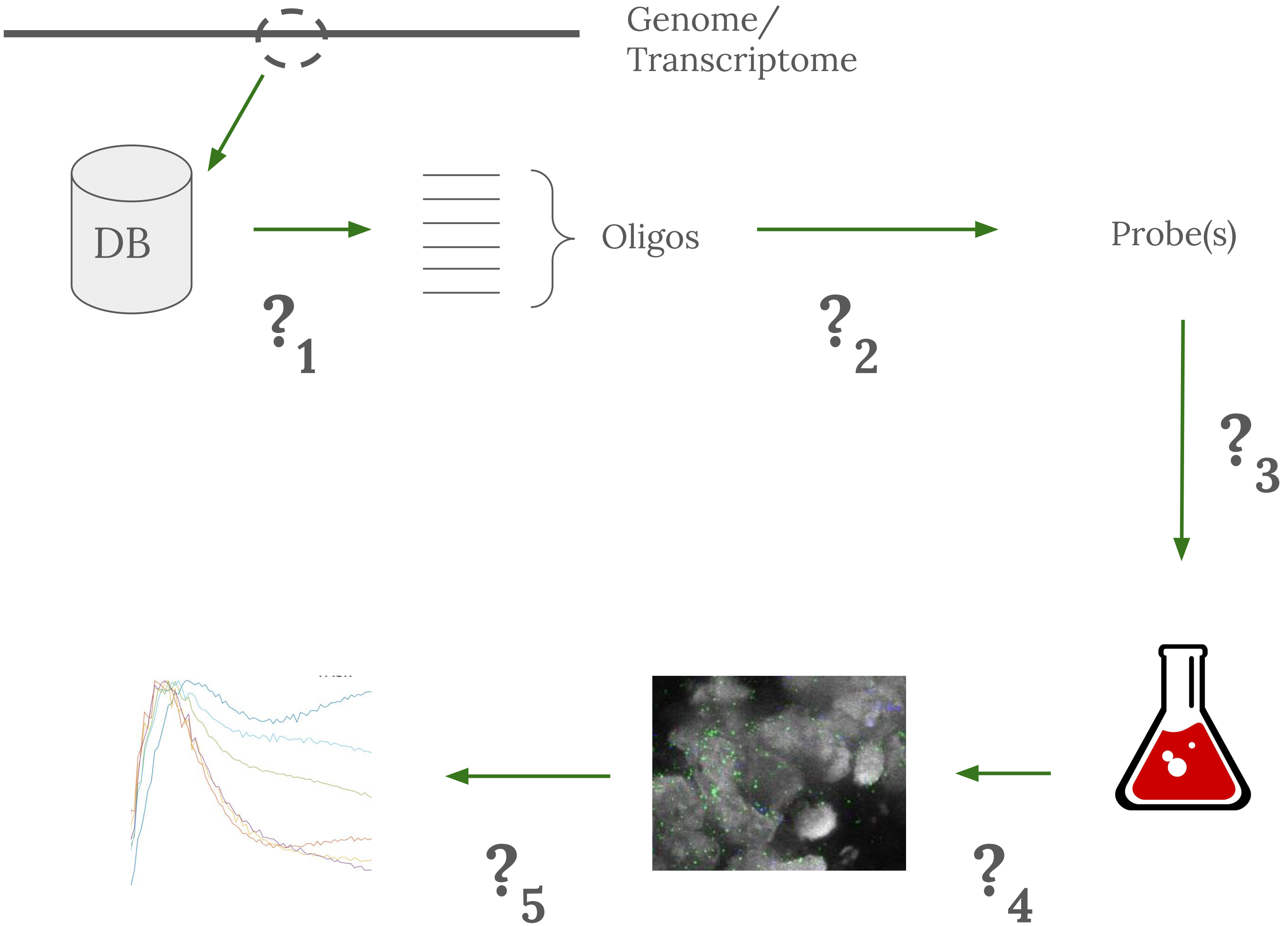
FLUORESCENCE IN SITU HYBRIDIZATION

FISH

From Theory to Practice

PROBES • IMAGING • DECONVOLUTION • ANALYSIS


Genome/
Transcriptome
DB
Oligos
Probe(s)


Genome/
Transcriptome
DB
Oligos
Probe(s)
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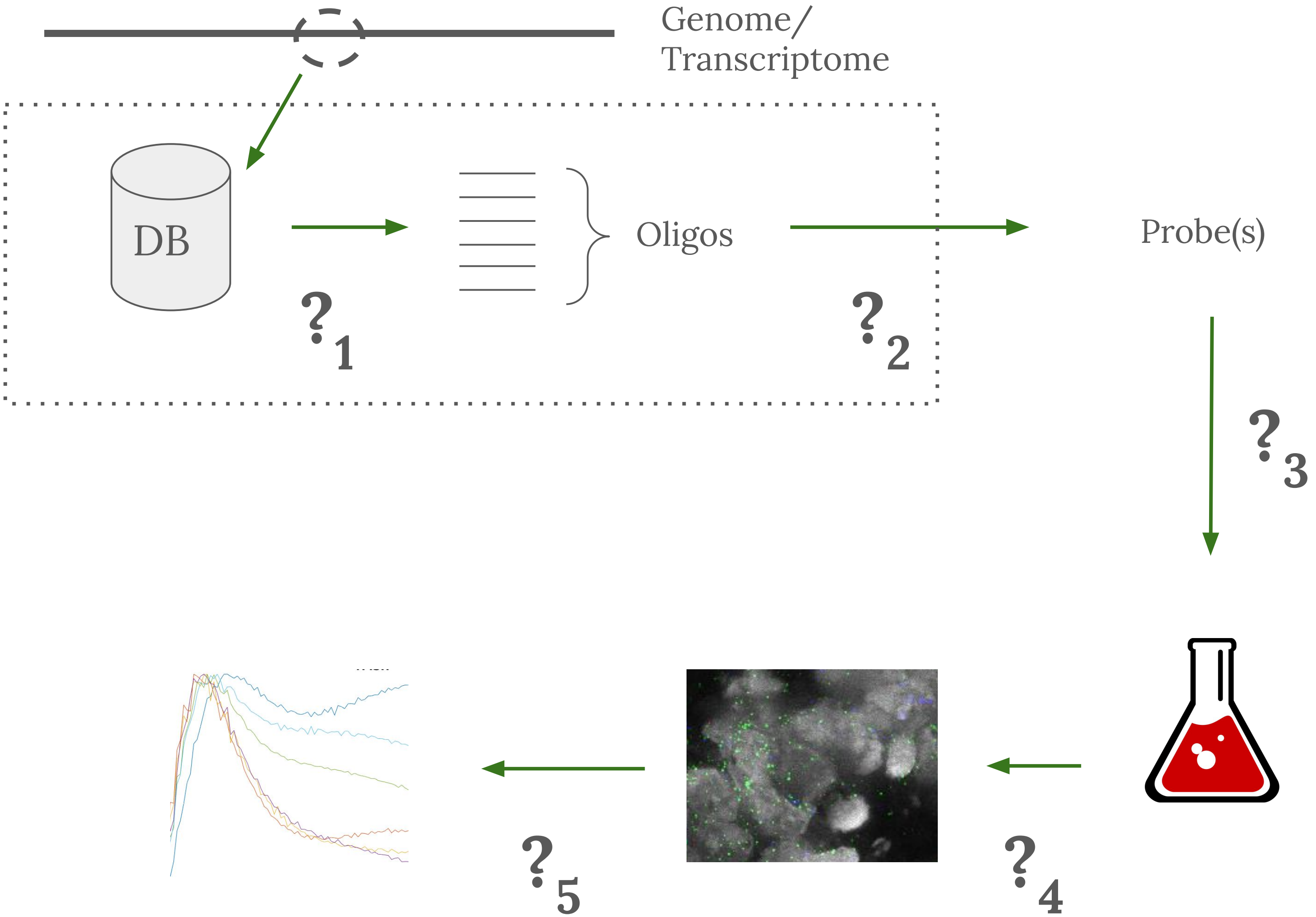

Genome/
Transcriptome
DB
Oligos
Probe(s)
?1
?2
?3
?4
?5

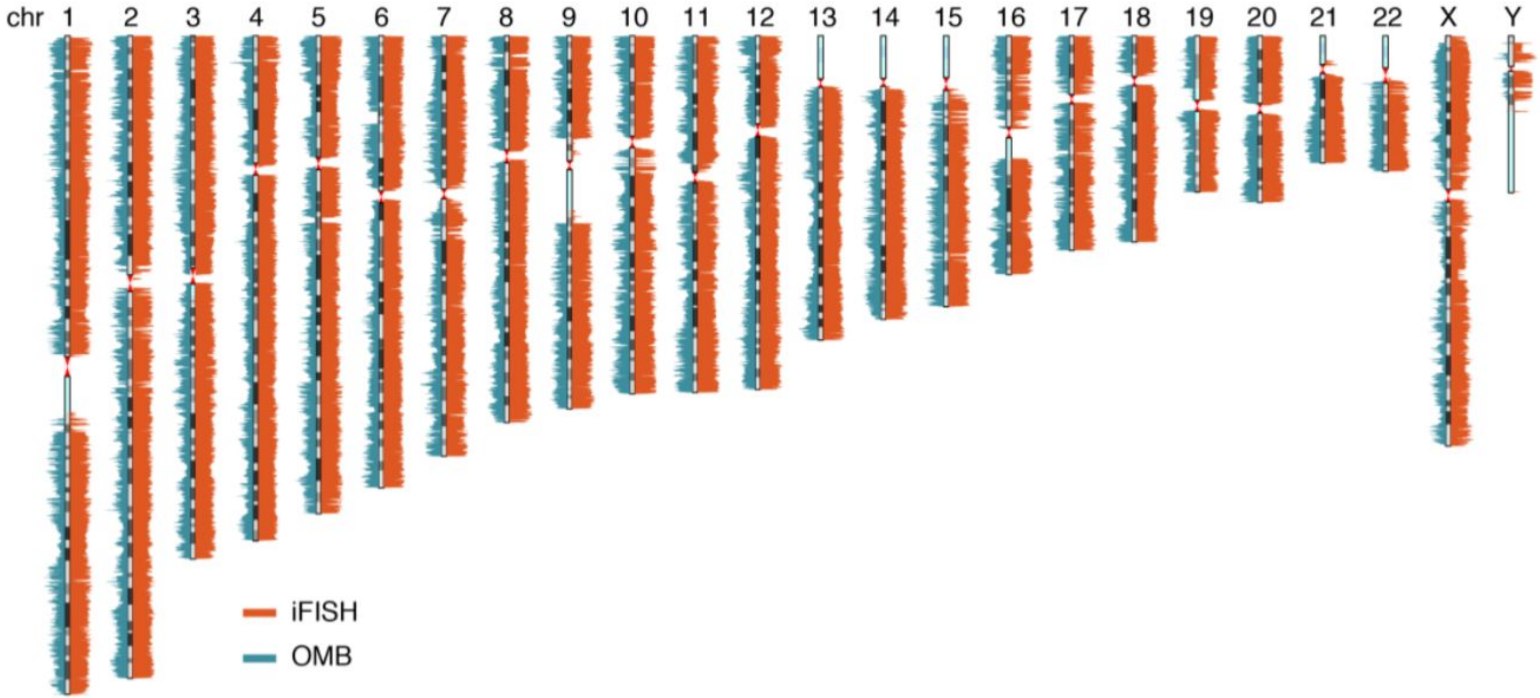
Article | Open access | Published: 09 April 2019

iFISH is a publically available resource enabling versatile DNA FISH to study genome architecture

Eleni Gelali, Gabriele Girelli, Masahiro Matsumoto, Erik Wernersson, Joaquin Custodio, Ana Mota, Maud Schweitzer, Katalin Ferenc, Xinge Li, Reza Mirzazadeh, Federico Agostini, John P. Schell, Fredrik Lanner, Nicola Crosetto & Magda Bienko

Nature Communications **10**, Article number: 1636 (2019) | Cite this article

Number of oligos per consecutive (non-overlapping) 100 kb genomic windows, in the 40-mers database (iFISH) and OligoMiner 'Balanced' hg19 (OMB) oligo database



Probe design software/ data bases

2006, **PROBER**, https://doi.org/10.1093/bioinformatics/btl273

2012–, **Stellaris RNA FISH Probe Designer**

2014–, **ProbeDesign**, https://github.com/arjunrajlaboratory/ProbeDesign

2018, **OligoMiner**, https://doi.org/10.1073/pnas.1714530115

2019, **iFISH**, https://doi.org/10.1038/s41467-019-09616-w

2020, **ProbeDealer**, https://doi.org/10.1038/s41598-020-76439-x

2021, **PaintSHOP**, https://doi.org/10.1038/s41592-021-01187-3

2025, **Oligostan-HT**, https://doi.org/10.1038/s41596-022-00750-2 (RNA)

2025, **TrueProbes**, https://doi.org/10.7554/eLife.108599.1 (RNA, preprint)

2026, **OligoMiner2**, https://github.com/beliveau-lab/OligoMiner2/ (code only)

Why not just iFISH/database XYZ and we are done?

Existing methods:

- RNA and or DNA?
- Which reference genome?
- Enough oligos where I need them?
- How long oligos?
- Validated for a scenario similar to mine?

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iFISH

- Designed for DNA
- Grch37/hg19
- Only 40-mers, non-overlapping

Current HT Pipeline

- Flexible :)

How are the databases constructed/ What makes a “good” oligo?

Oligo selection

?

=

Sequence alignment

anti-aligner

A) Maximize on-target binding probability
B) Minimize off-target bindings

aligner

A) Find <u>the unique</u> origin of a fragment.
B) Avoid being fooled by sequencing errors and SNVs.

iFISH: bwa
Oligominer: bowtie2
Current HT: HUSH+?

Why not use blast manually?

AAGCAGAAATGAACTCTACGGTTT
TAAGTCGGAACTAAAGGTAACAAA
ATACTCATGATA (n=60)

Sequences producing significant alignments

Download Select columns Show 100

select all 1 sequences selected

GenBank Graphics Distance tree of results MSA Viewer

	Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
Genomic sequences									
☑	Homo sapiens chromosome 11, GRCh38.p14 Primary Assembly	Homo sapiens	109	109	100%	1e-22	100.00%	135086622	NC_000011.10

Sequences producing significant alignments Download Select columns Show 100

select all 1 sequences selected GenBank Graphics Distance tree of results MSA Viewer

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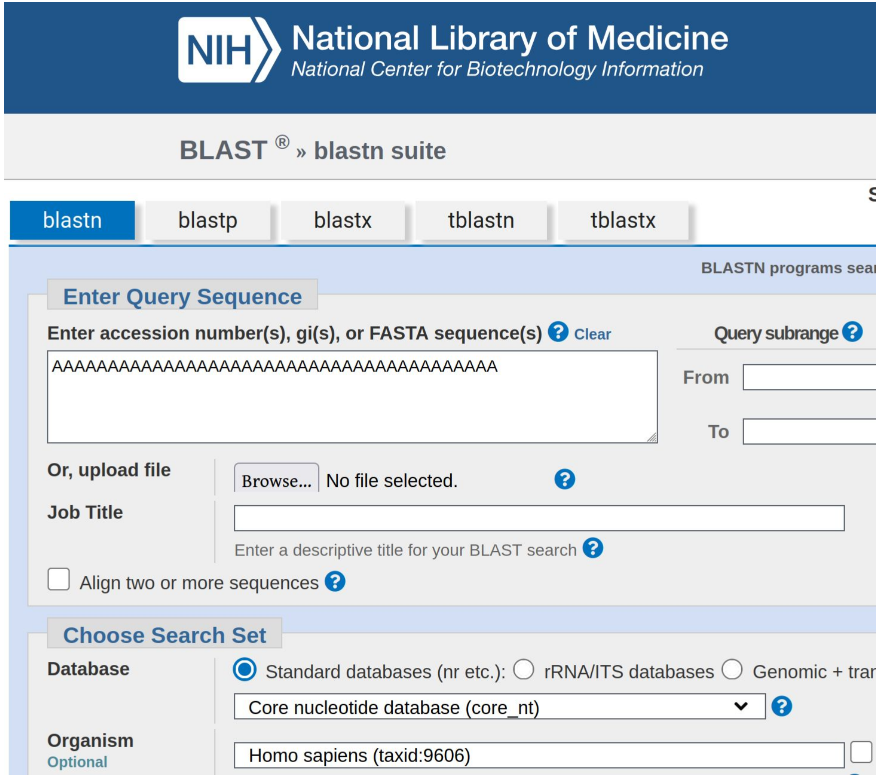
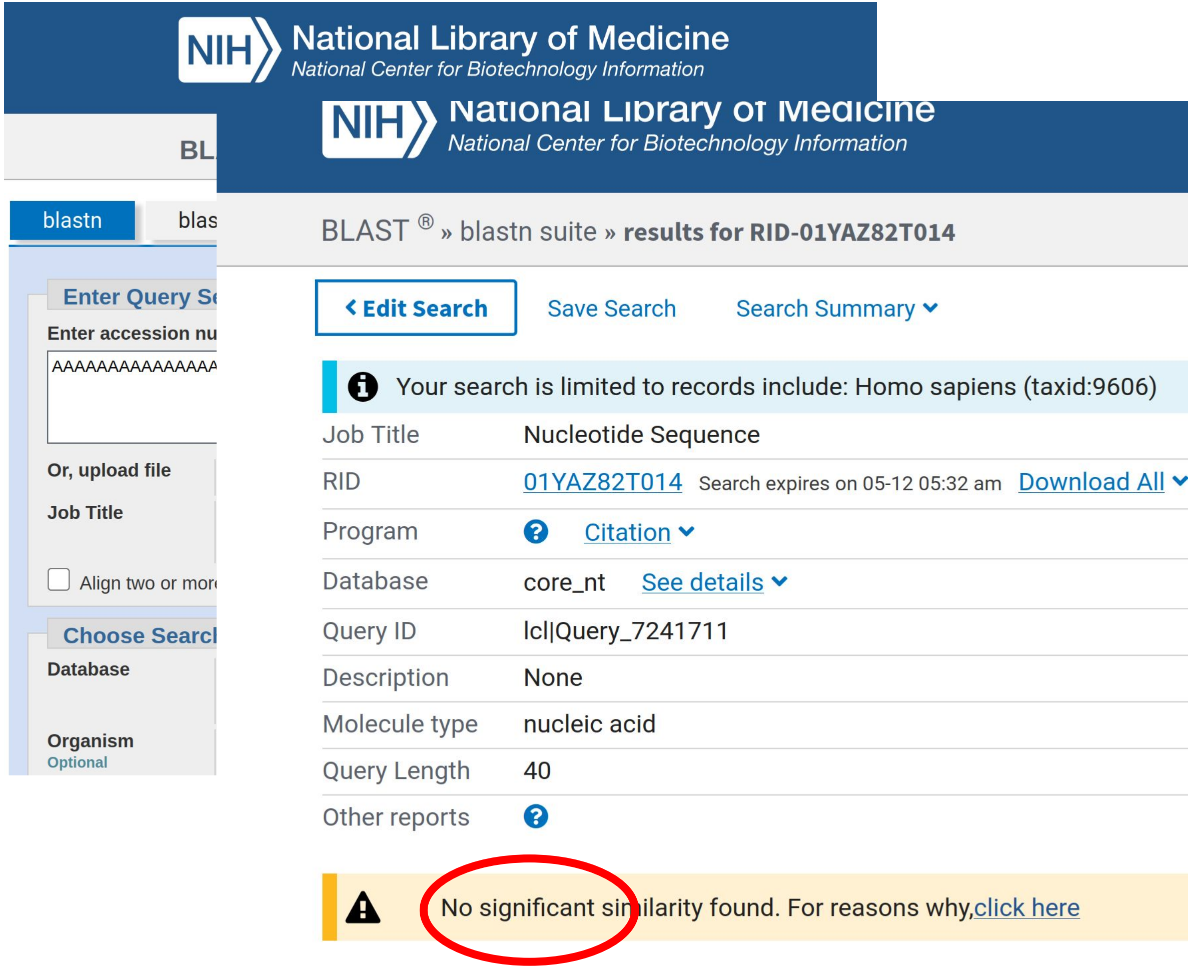
☑	Homo sapiens chromosome 11, GRCh38.p14 Primary Assembly	Homo sapiens	109	109	100%	1e-22	100.00%	135086622	NC_000011.10

```
GGATTATAGGCATGTGCCACCACACCCGGCTAATTTTTTTGTATTTTTAGTAGAGACGGG NC_000001.11 Homo sapiens chromosome 1, GRCh38.p14 Primary Assembly:39938608:39938668
||||||||||||| ||||||||||||||||||||||||||||||||||| |||||||||| fHD=2.00, match=97%
GGATTATAGGCATCTGCCACCACACCCGGCTAATTTTTTTGTATTTTTGGTAGAGACGGG query, RC
```

```
Hamming distance, number of hits
  0, 1
  1, 0
  2, 1
  3, 12
  4, 74
  5, 419
…
```


NIH
National Library of Medicine
National Center for Biotechnology Information
BLAST ® » blastn suite
blastn
blastp
blastx
tblastn
tblastx
Enter Query Sequence
Enter accession number(s), gi(s), or FASTA sequence(s)
Clear
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
Query subrange
From
To
Or, upload file
Browse... No file selected.
Job Title
Enter a descriptive title for your BLAST search
Align two or more sequences
Choose Search Set
Database
Standard databases (nr etc.):
rRNA/ITS databases
Core nucleotide database (core_nt)
Organism
Optional
Homo sapiens (taxid:9606)



National Library of Medicine
National Center for Biotechnology Information
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National Center for Biotechnology Information
BLAST ® » blastn suite » results for RID-01YAZ82T014
Edit Search
Save Search
Search Summary
Your search is limited to records include: Homo sapiens (taxid:9606)
Job Title
Nucleotide Sequence
RID
01YAZ82T014
Search expires on 05-12 05:32 am
Download All
Program
Citation
Database
core_nt
See details
Query ID
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Description
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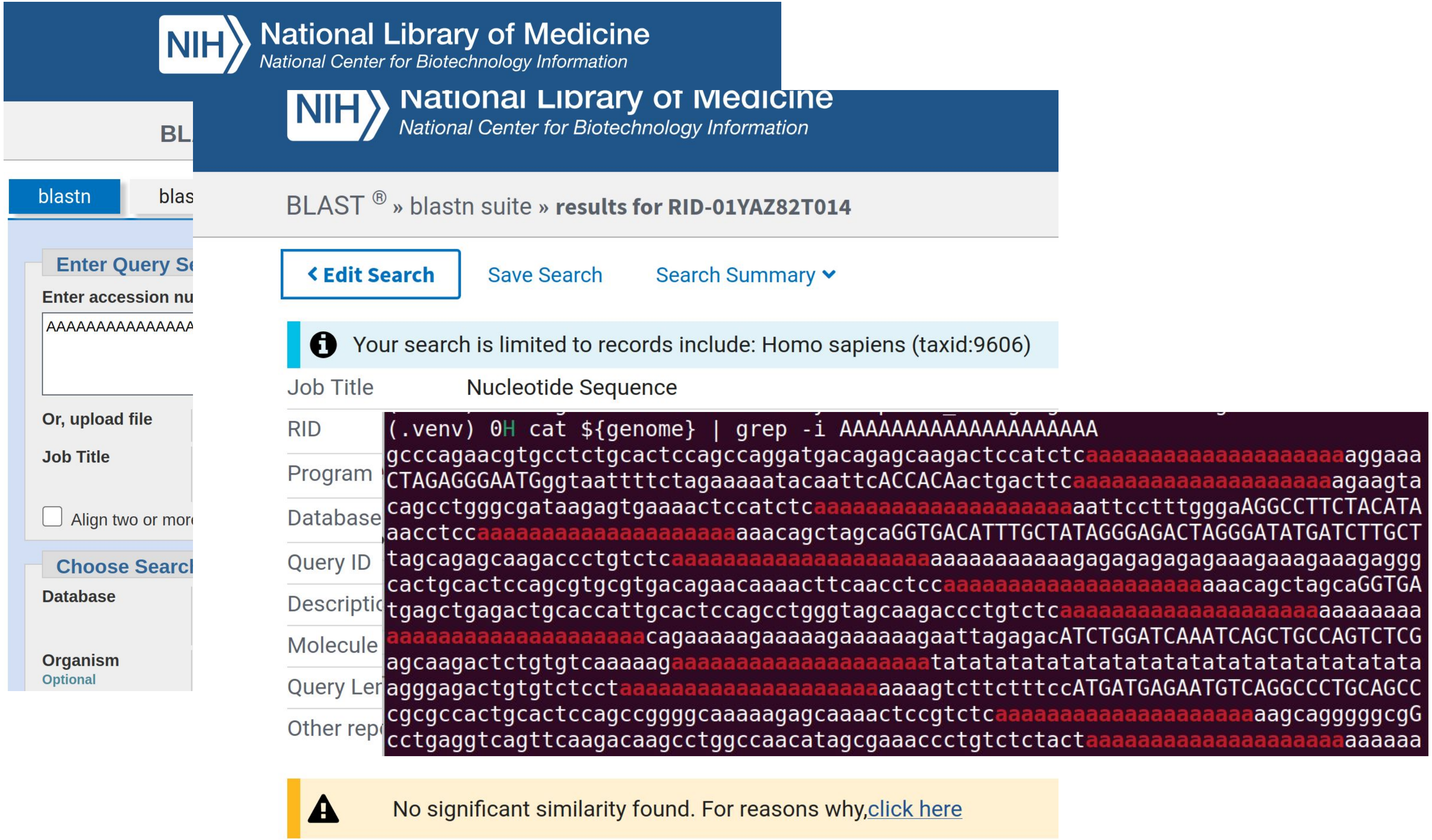
Molecule type
nucleic acid
Query Length
40
Other reports
No significant similarity found. For reasons why,click here



NIH National Library of Medicine
National Center for Biotechnology Information
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AAAAAAAAAAAAAAA
Or, upload file
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Choose Search
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NIH National Library of Medicine
National Center for Biotechnology Information
BLAST ® » blastn suite » results for RID-01YAZ82T014
Edit Search
Save Search
Search Summary
Your search is limited to records include: Homo sapiens (taxid:9606)
Job Title
Nucleotide Sequence
RID
Program
Database
Query ID
Descriptio
Molecule
Query Len
Other repo
(.venv) 0H cat ${genome} | grep -i AAAAAAAAAAAAAAAAAAAA
gcccagaacgtgcctctgcactccagccaggatgacagagcaagactccatctcaaaaaaaaaaaaaaaaaaaaaggaaa
CTAGAGGGAATGggtaatttctagaaaatacaattcACCACAactgacttcaaaaaaaaaaaaaaaaaaaaagaagta
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cgcgccactgcactccagccggggcaaaaagagcaaaactccgtctcaaaaaaaaaaaaaaaaaaaaaagcagggggcgG
cctgaggtcagttcaagacaagcctggccaacatagcgaaaccctgtctctactaaaaaaaaaaaaaaaaaaaaaaaaaa
No significant similarity found. For reasons why,click here

Melting temperature / binding probability

T_m

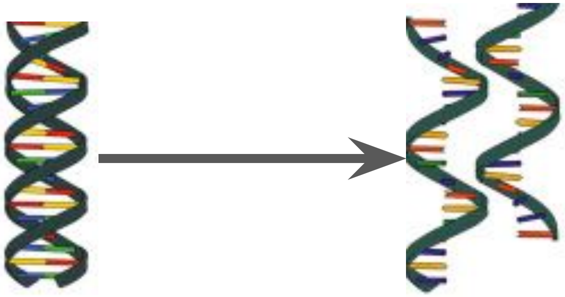
When there is a 50% probability of ds and 50% probability of ss.

Predictions/calculations based on experimental data.

Double strand to single strands

Denaturation

dsDNA/dsRNA -> ss.



Adopted from https://doi.org/10.1016/j.polymertesting.2019.106158

Single strands to double strand: **competitive mechanism I**

Hybridization

Pairing of strands from different sources

Renaturation

The reverse of denaturation

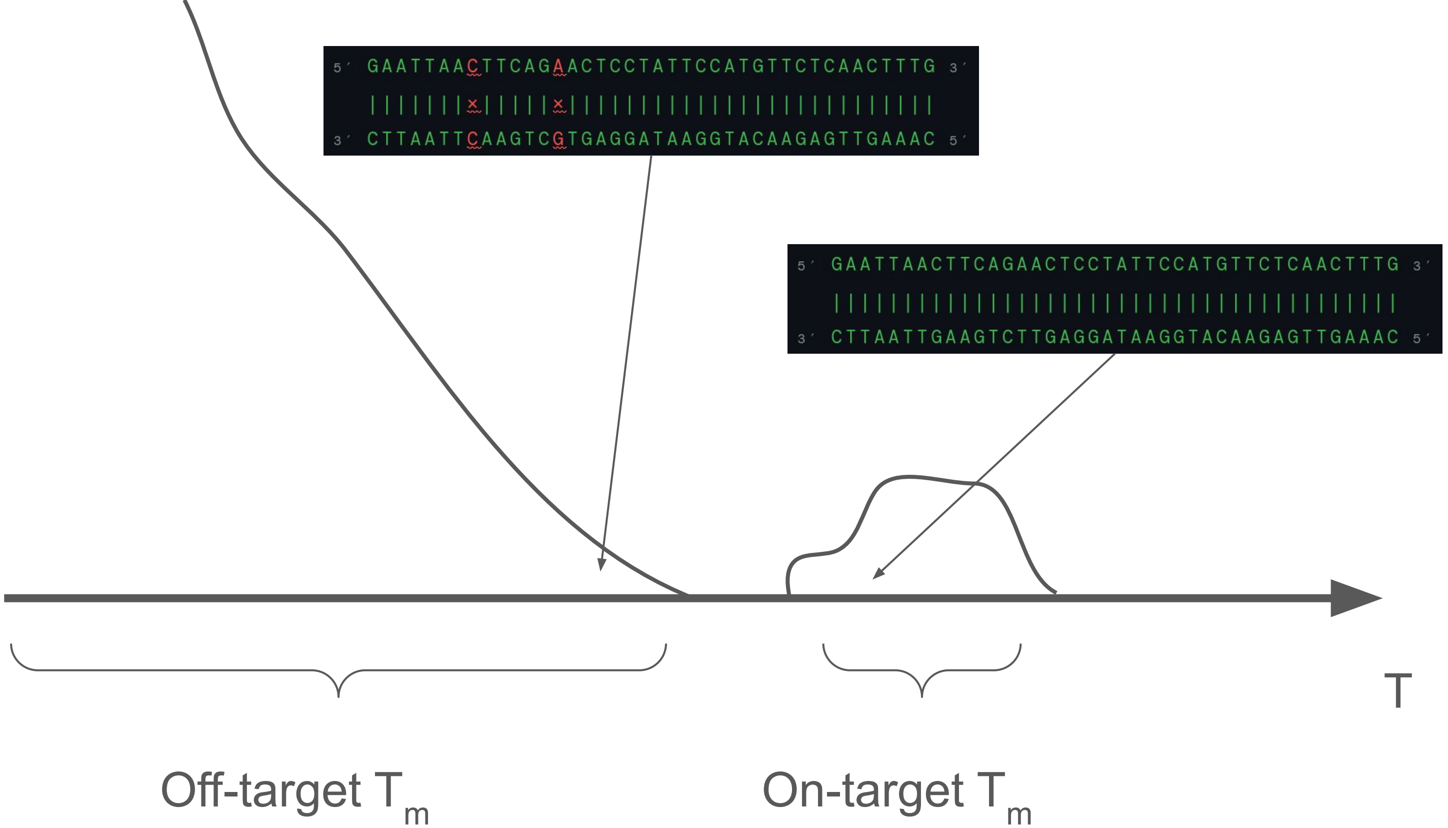
On-target vs off-target hybridization: **competitive mechanism II**

On-target bindings

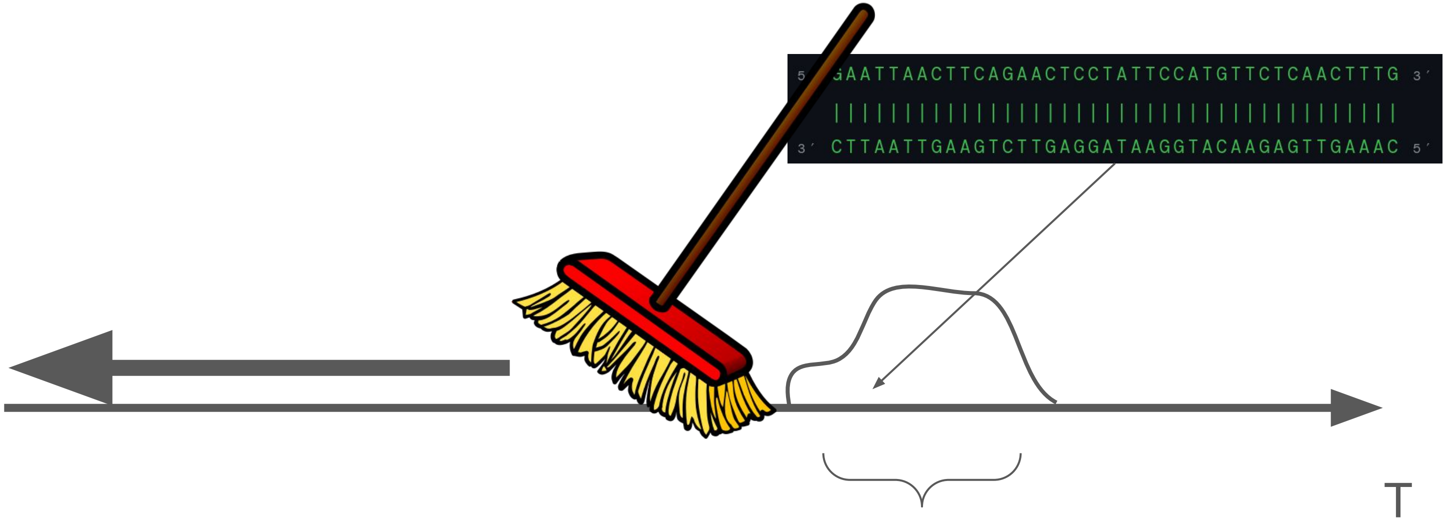
Hybridization between complementary DNA/RNA.

Off-target bindings

Hybridization between non-complementary DNA/RNA.


5′ GAATTAACTTCAGAACTCCTATTCCATGTTCTCAACTTTG 3′
3′ CTTAATTCAAGTCGTGAGGATAAGGTACAAGAGTTGAAAC 5′
5′ GAATTAACTTCAGAACTCCTATTCCATGTTCTCAACTTTG 3′
3′ CTTAATTGAAGTCTTGAGGATAAGGTACAAGAGTTGAAAC 5′
T
Off-target Tm
On-target Tm

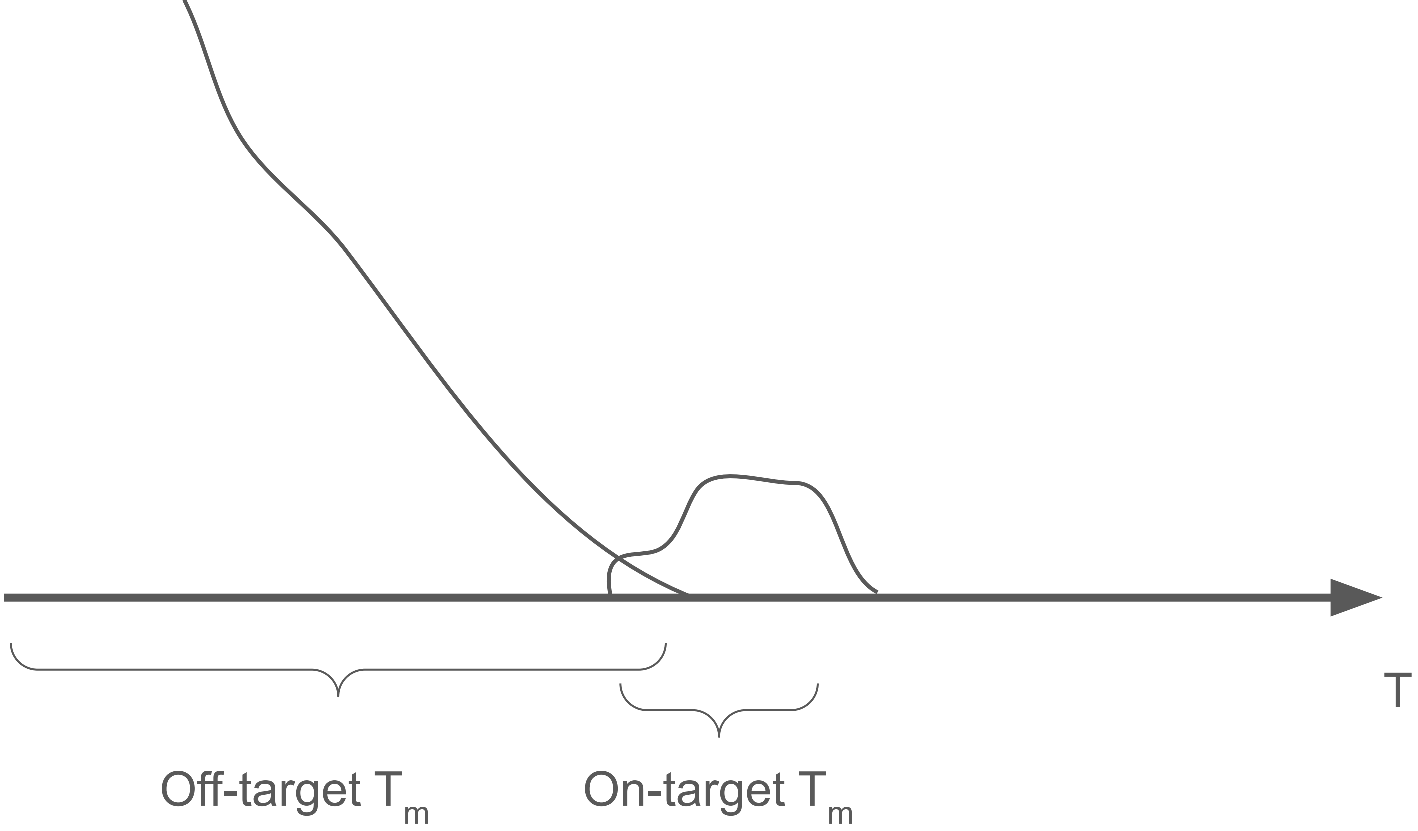
Wash step



On-target T_m

Removes:

- non-bound
- weakly bound
  electrostatically bound
- …



T
Off-target T_m
On-target T_m

Make ΔT_m high enough!?

T_m From basic counts

1961, Marmur & Doty

$$T_m = 69.3 + 0.41\,([CG])$$

1986, Thein & Wallace

$$T_m = 2(A + T) + 4(C + G)$$

T_m From “stacking”

that is the temperature of the human body, and get worse as the temperature deviates from 50°C (see References 71 and 72 for discussion of error extrapolation). The two-state melting temperature (T_M) may be calculated with Equation 3:

$$T_M = \Delta H^\circ \times 1000/(\Delta S^\circ + R \times \ln(C_T/x)) - 273.15, \qquad 3.$$

TABLE 1 Nearest-neighbor thermodynamic parameters for DNA Watson-Crick pairs in 1 M NaCl[a]

Propagation sequence	ΔH° (kcal mol^{-1})	ΔS° (e.u.)	ΔG°_{37} (kcal mol^{-1})
AA/TT	−7.6	−21.3	−1.00
AT/TA	−7.2	−20.4	−0.88
TA/AT	−7.2	−21.3	−0.58
CA/GT	−8.5	−22.7	−1.45
GT/CA	−8.4	−22.4	−1.44
CT/GA	−7.8	−21.0	−1.28
GA/CT	−8.2	−22.2	−1.30
CG/GC	−10.6	−27.2	−2.17
GC/CG	−9.8	−24.4	−2.24
GG/CC	−8.0	−19.9	−1.84
Initiation	+0.2	−5.7	+1.96
Terminal AT penalty	+2.2	+6.9	+0.05
Symmetry correction	0.0	−1.4	+0.43

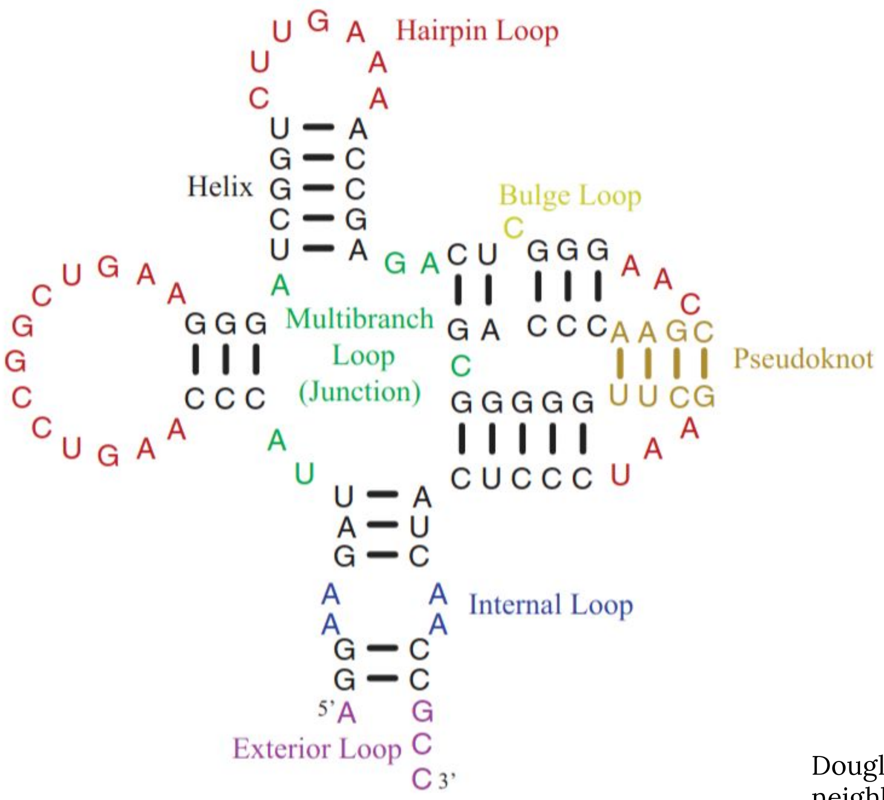
Review > Annu Rev Biophys Biomol Struct. 2004:33:415-40.
doi: 10.1146/annurev.biophys.32.110601.141800.

The thermodynamics of DNA structural motifs

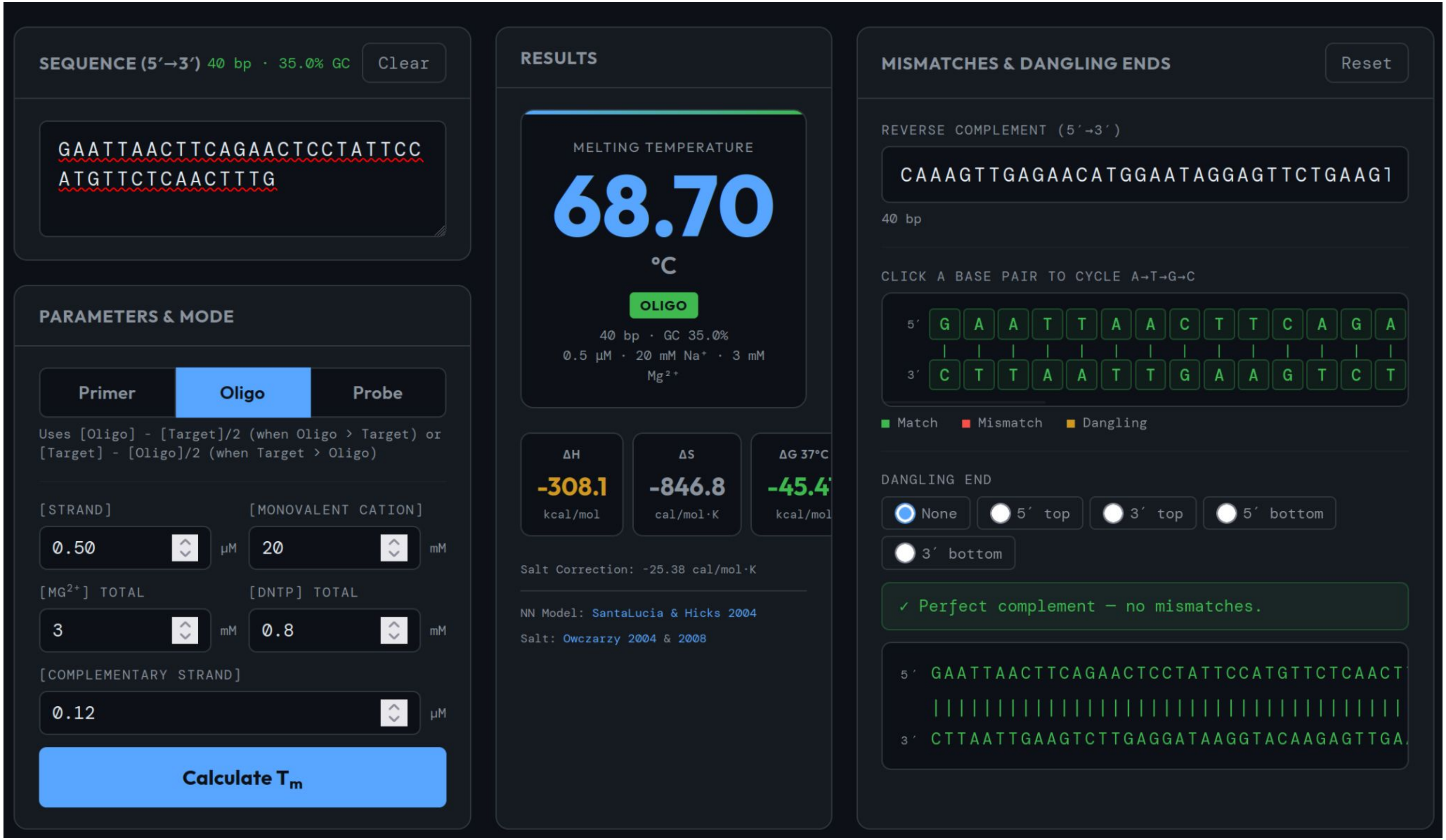
John SantaLucia Jr [1], Donald Hicks

Affiliations + expand

PMID: 15139820 DOI: 10.1146/annurev.biophys.32.110601.141800



Douglas H. Turner, David H. Mathews, NNDB: the nearest neighbor parameter database for predicting stability of nucleic acid secondary structure, *Nucleic Acids Research*, Volume 38, Issue suppl_1, 1 January 2010, Pages D280–D282, https://doi.org/10.1093/nar/gkp892



https://api.dna-utah.org/tmtool/

For oligo section …

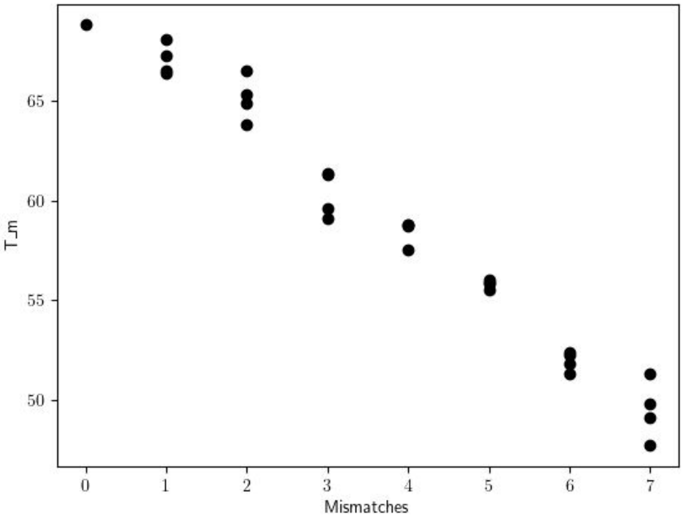
GC content – can work as a first approximation for Tm

20-mer

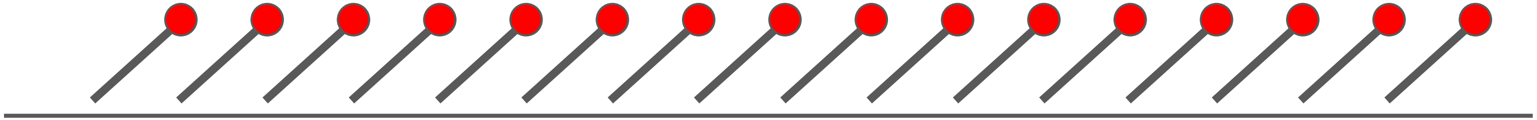
- 100% G(C) -> 87.01 °C
- 100% A(T) -> 45.25 °C

For off-targets: <u>Mismatches</u> and indels decrease T_m

Separate strategies for long vs short oligos

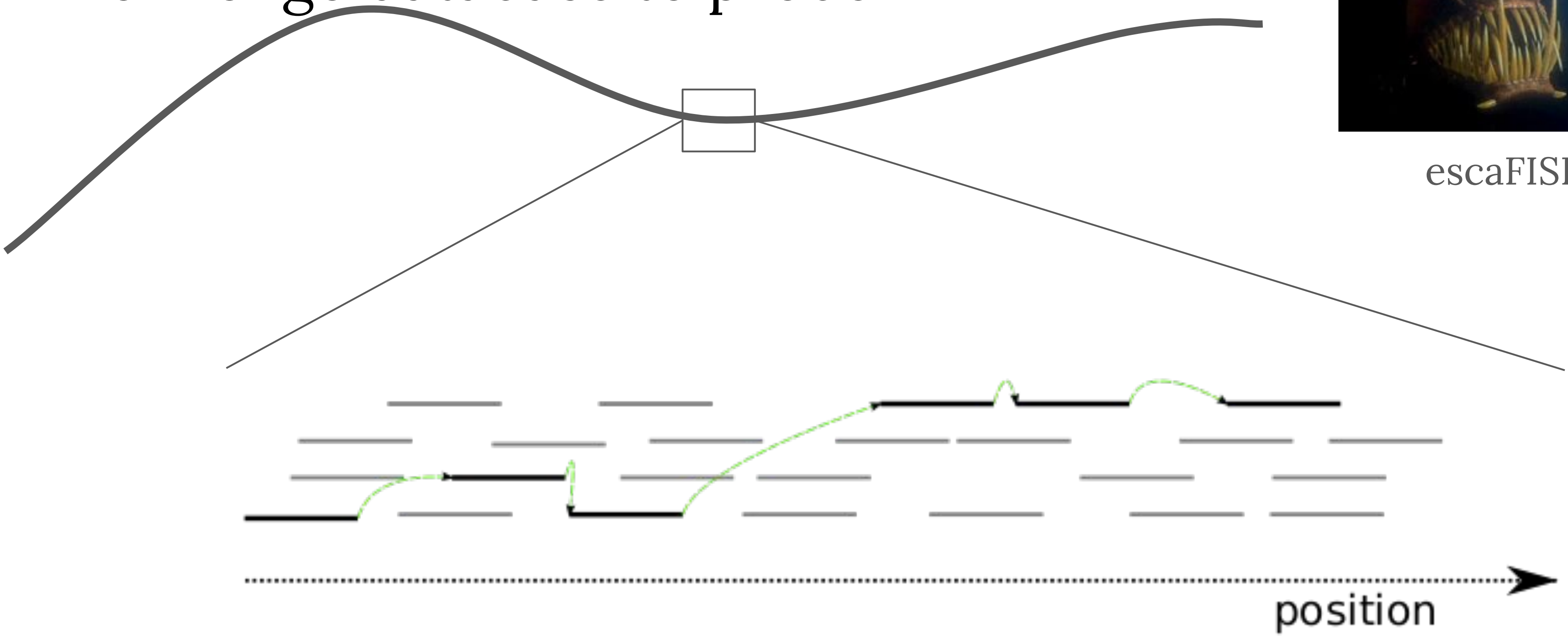


Introducing mismatches to GAATTAACTTCAGAACTCCTATTCCATGTTCTCAACTTTG



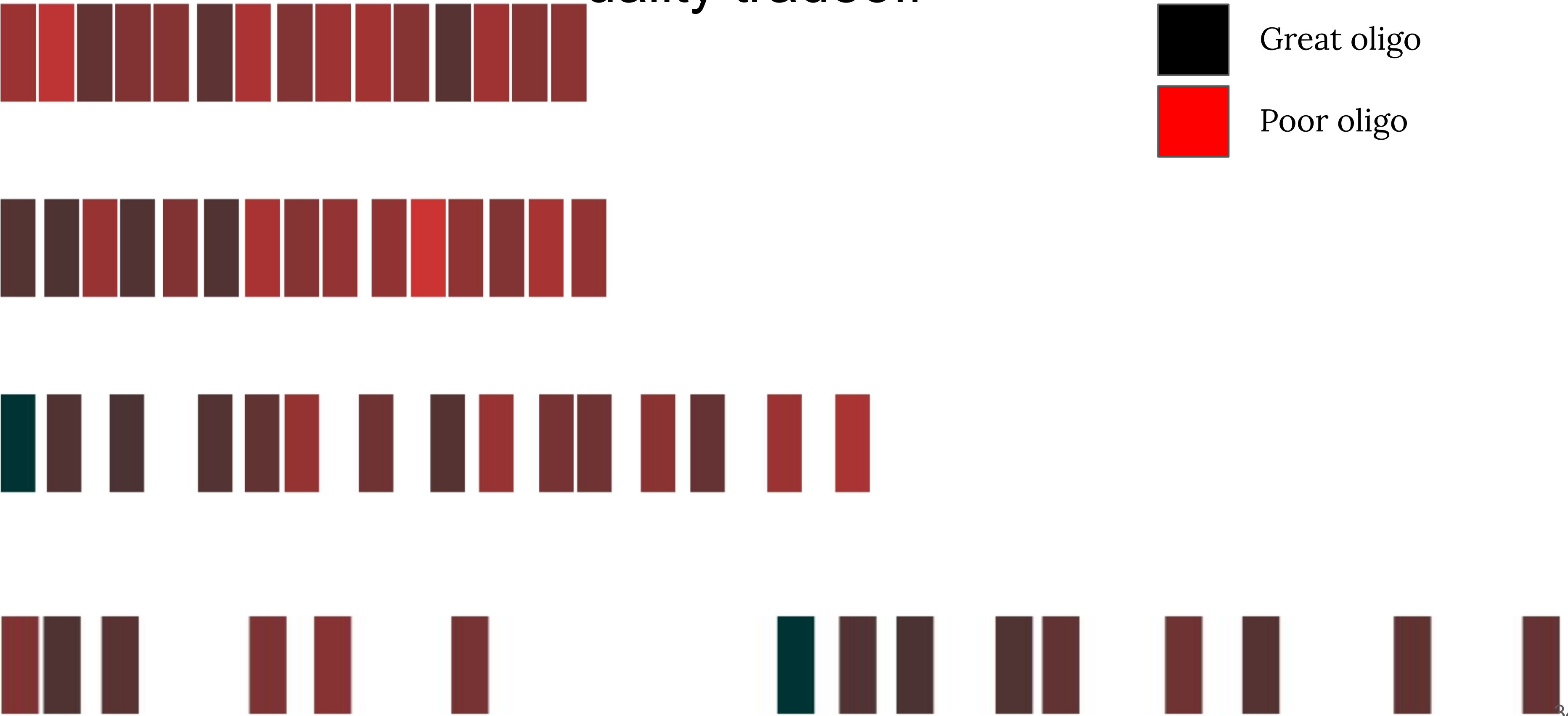
Building one probe from many oligos

From oligo database to probe



escaFISH

uality tradeoff



Unique oligos / Comparing oligos

Hamming distance

= The number of mismatching/non Crick-Watson base pairs for equal-length oligos.

```
GATTTAGCTTCAGAAGTCCTAGA NC_060925.1 Homo sapiens isolate CHM13 chromosome 1, alternate assembly T2T-CHM13v2.0:209435802:209435825
|| ||| |||||||| |||||   fHD=5.00, match=78%
GAATTAACTTCAGAACTCCTATC query

TAATTTACTTCAGAATTCCTATA NC_060931.1 Homo sapiens isolate CHM13 chromosome 7, alternate assembly T2T-CHM13v2.0:141650194:141650217
 |||| ||||||||| ||||||  fHD=4.00, match=83%
GAATTAACTTCAGAACTCCTATC query
```

Levenshtein distance

The number of edits required to match two strings.

```
levenshtein=2
GAATTAACTTCAGAACTCCTATC (query)
|||||||||||||| |||||| |
GAATTAACTTCAGATCTCCTAAC (ref)
```

Also Hamming distance = 2

```
levenshtein=2
GAATTAACTTCAG--AACTCCTATC (query)
|||||||||||||  ||||||||||
GAATTAACTTCAGTTAACTCCTATC (ref)
```

Indel of length 2

```
levenshtein=6
GAATTAACTTC--AGAACTCCCCTATC (query)
|||||||||||   |||||  |||| |
GAATTAACTTCTGTGAACT--CCTAAC (ref)
```

2x2 indels + 2 mm

Vs genome – back-of-the-envelope calculations

Score	Expect	Identities	Gaps	Strand
75.0 bits(40)	3e-12	40/40(100%)	0/40(0%)	Plus/Plus

```
Query  1         GAGAACCTTCCGACGGAAGTGACGTCGTTACGACGCGCGA  40
                 ||||||||||||||||||||||||||||||||||||||||
Sbjct  62943255  GAGAACCTTCCGACGGAAGTGACGTCGTTACGACGCGCGA  62943294
```

1 sequence
$T = G \times L \,/\, F \cong L$ s

All possible sequences
$T = G \times G \times L \,/\, F \cong ?$ s

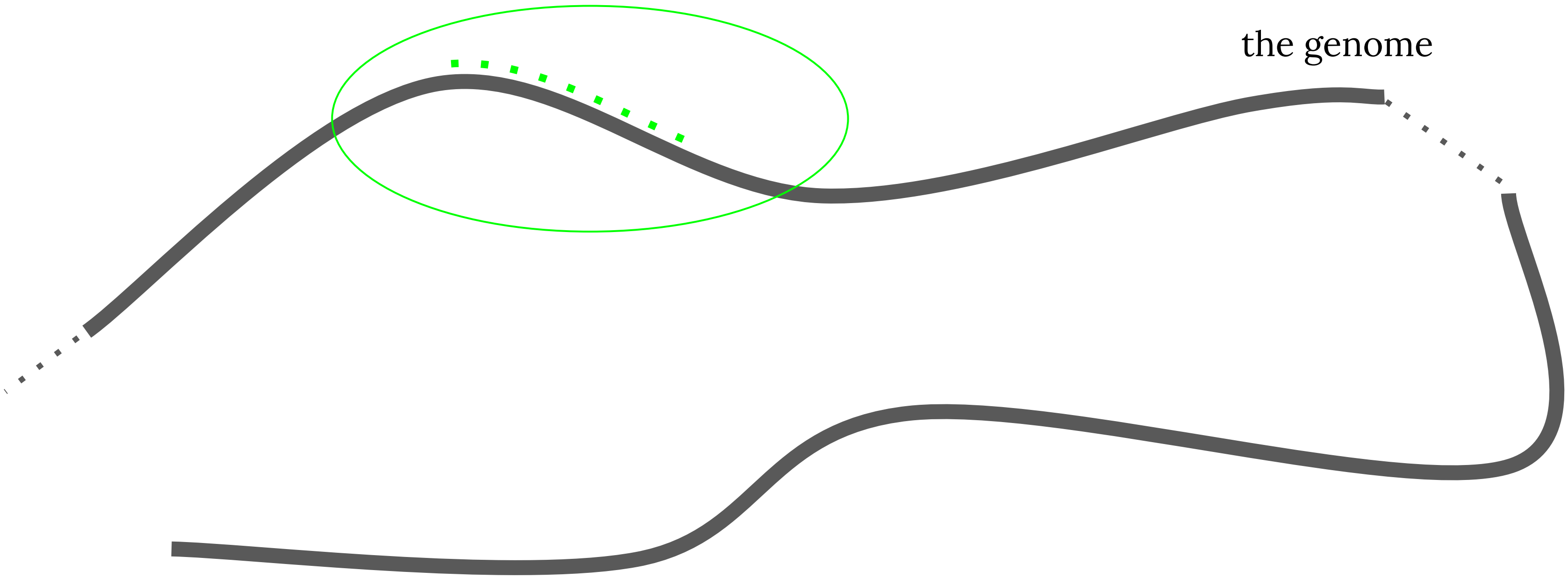
- Genome size
  - $G \cong 3 \times 10^9$
- Sequence length
  - $L = 40$
- CPU Floating Point Operations s^{-1}
  - $F \cong 3 \times 10^9$

All vs all

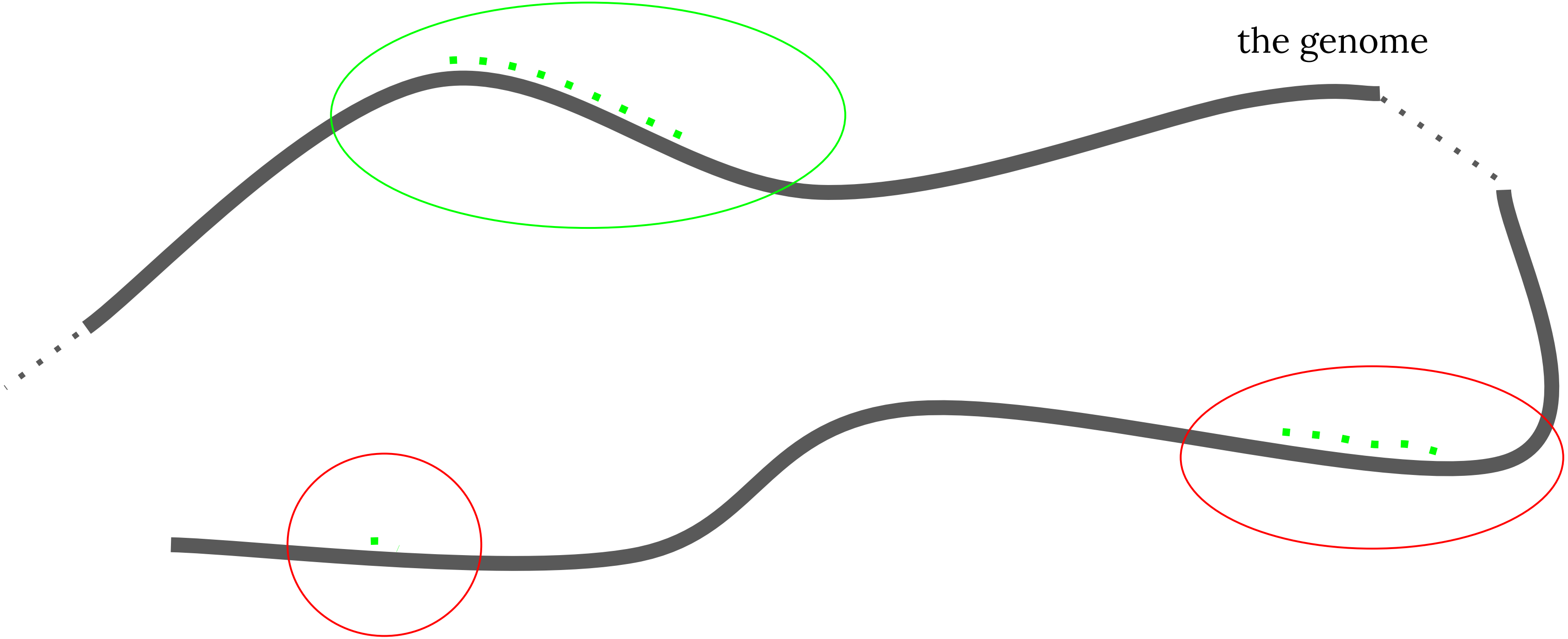
$$T = G \times G \times L \;/\; F$$
$$\cong 120 \times 10^9 \text{ s}$$
$$\cong 4{,}000 \text{ years}$$

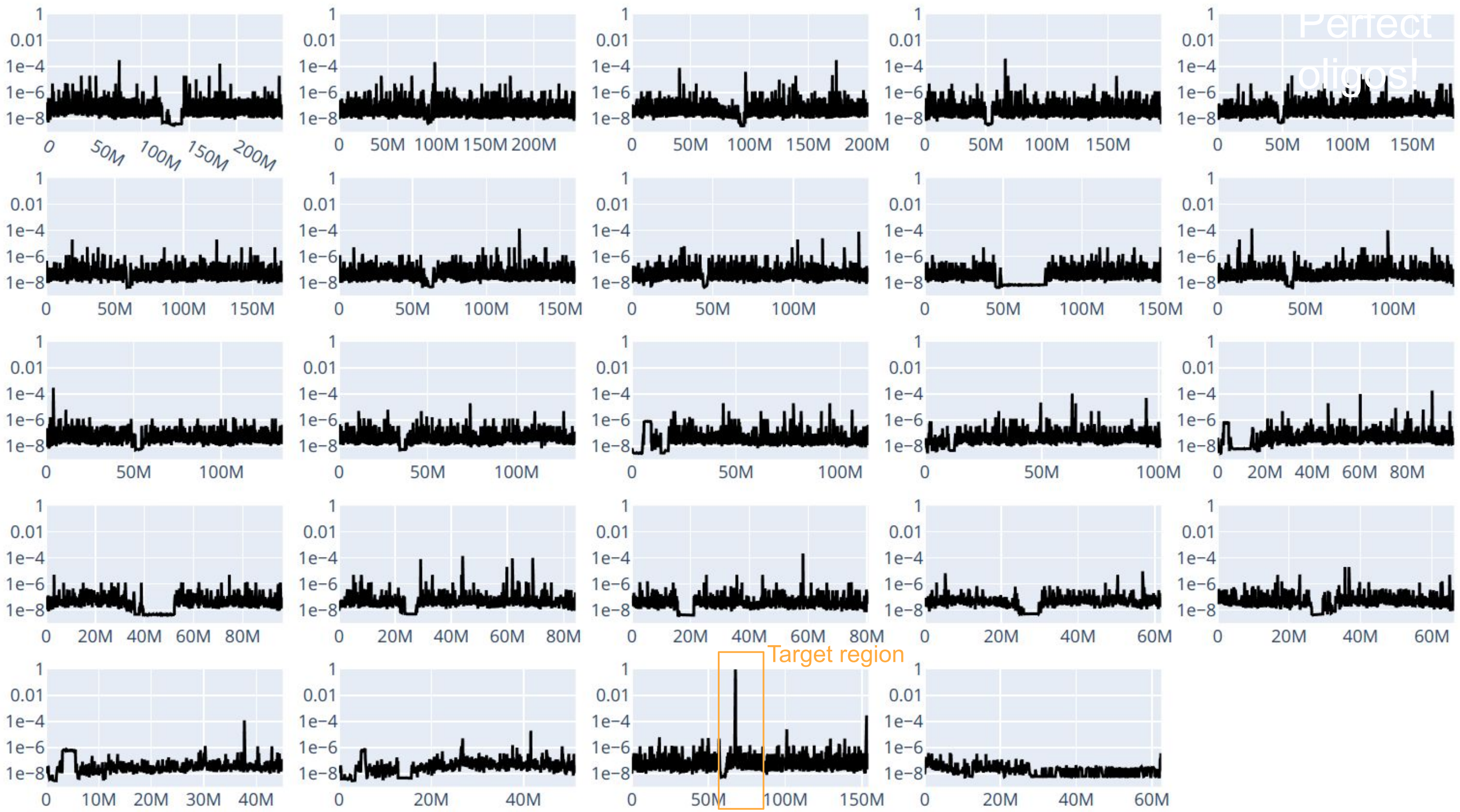
Some last probe considerations

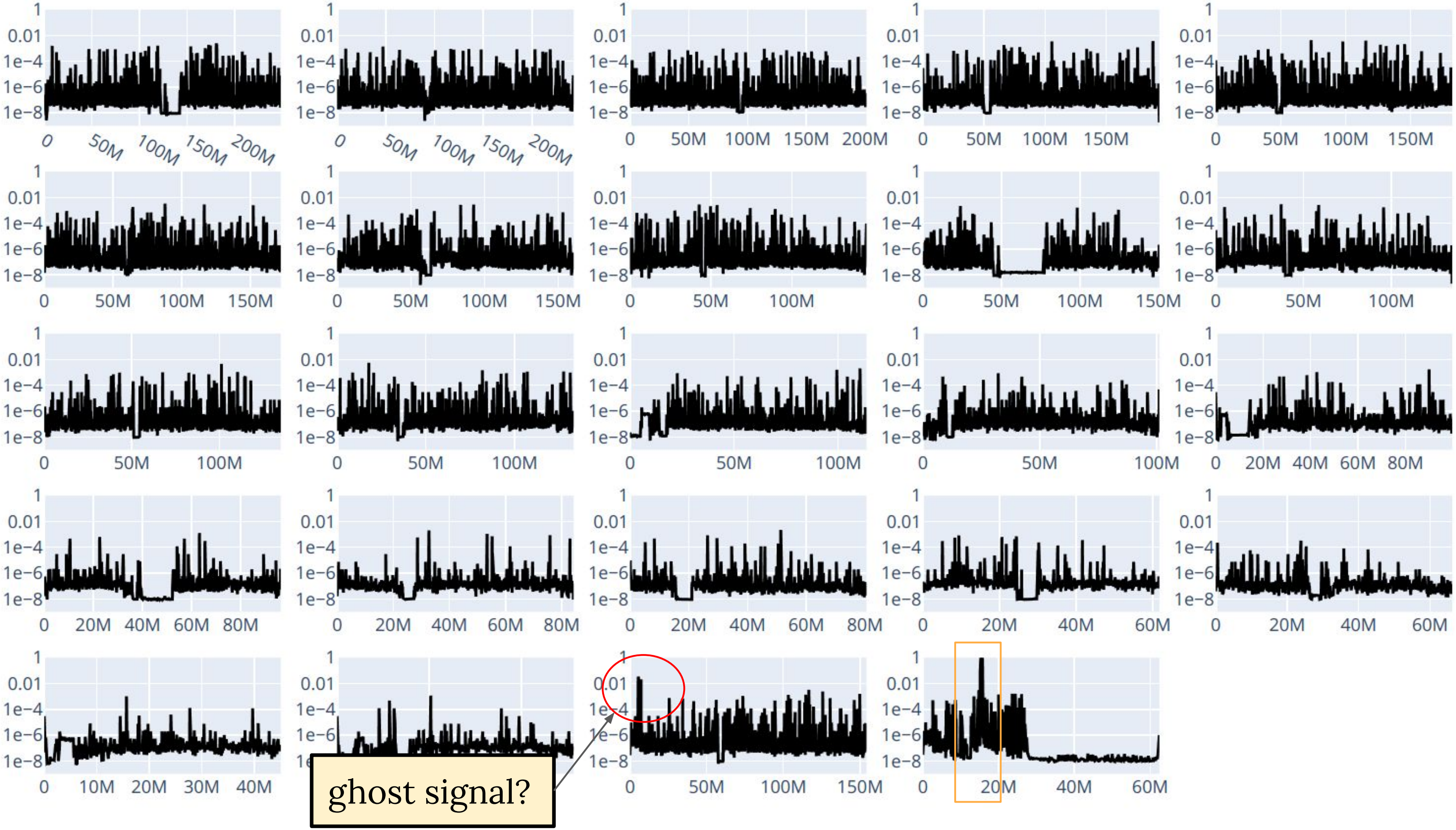
Co-localization of off-targets?



Co-localization of off-targets?




Perfect oligos!
Target region


1
0.01
1e−4
1e−6
1e−8
0 50M 100M 150M 200M
0 50M 100M 150M 200M
0 50M 100M 150M 200M
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0 20M 40M 60M
0 20M 40M 60M
0 10M 20M 30M 40M
0 50M 100M 150M
0 20M 40M 60M
ghost signal?